3	**(a)**	$2k/m = \omega^2$	M1	
		$\omega = 2\pi f$	M1	
		$(2 \times 64 / 0.810) = (2\pi \times f)^2$ leading to $f = 2.0$ Hz	A1	[3]
	(b)	$v_0 = \omega x_0$ *or* $v_0 = 2\pi f x_0$ *or* $v = \omega(x_0^2 - x^2)^{1/2}$ <u>and</u> $x = 0$	C1	
		$v_0 = 2\pi \times 2.0 \times 1.6 \times 10^{-2}$		
		$= 0.20\,\mathrm{m\,s^{-1}}$	A1	[2]
	(c)	frequency: reduced/decreased	B1	
		maximum speed: reduced/decreased	B1	[2]